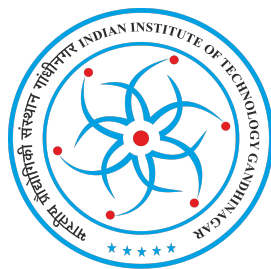


INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



Assignment Report

M.Tech in Integrated Circuit Design and Technology (ICDT)

Name:	Kalaparthi Vikas - 25210064
Subject Code:	ES 626
Subject Name:	Microfabrication and Semiconductor Processes
Assignment No:	03
Submission Date:	19 November 2025

Question 1

Based on the dimensions provided in the layout file and the process flow and device parameters available at the given link, simulate and construct the single-fin FinFET structure depicted in Figure 1. Analyze the resulting structure and provide detailed answers to the subsequent questions:

1. What is the maximum activation anneal temperature we can use to prevent dopants from diffusing beyond the Fin height and to maintain a channel length of at least 25 nm? (Take tilt angle as zero, Fin height is the height of the fin structure until the start of STI oxide.)

Answer: Maximum activation anneal — I chose 1100°C

2. Keeping the above activation anneal parameters constant, what is the allowable tilt angle during ion implantation to ensure proper doping in the Fins?

Answer: Allowable tilt angle — I chose 8.5°

3. What is the shadowing effect? Plot the dopant distribution across Fin width for different dose, energy, and tilt.

Shadowing Effect

Shadowing in ion implantation arises when the incident ion beam is partially blocked by the fin sidewalls, the gate stack, or the spacer oxide. As a result, ions cannot uniformly reach all regions of the fin, particularly at non-zero tilt angles. This leads to an asymmetric dopant distribution with higher concentration on the beam-facing side of the fin and reduced concentration on the shadowed side.

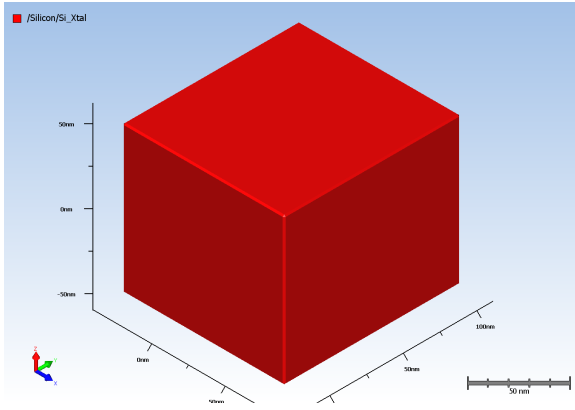


Figure 1: Wafer Setup

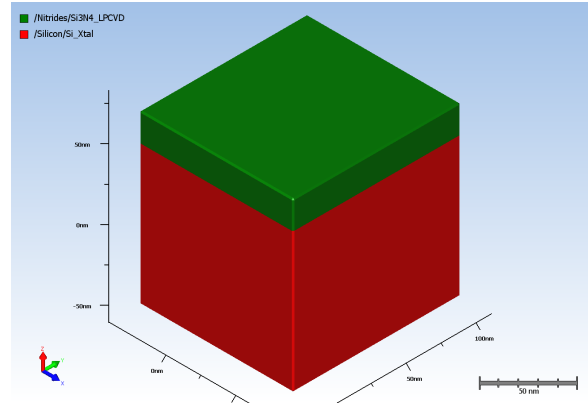


Figure 2: SiN Hardmask

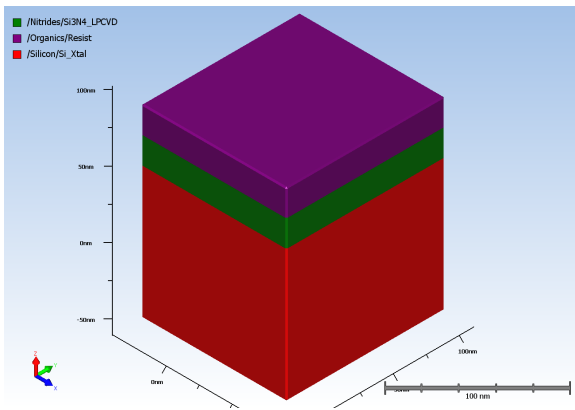


Figure 3: Resist for SiN

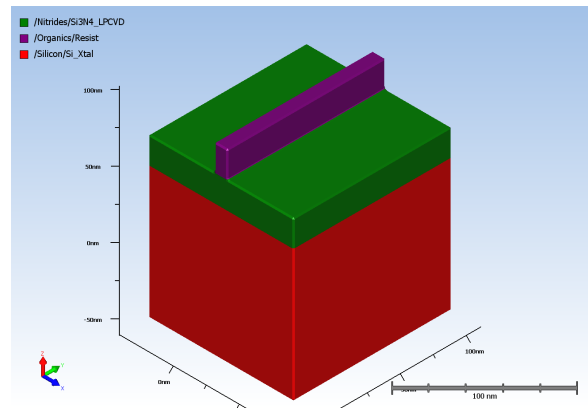


Figure 4: Expose resist

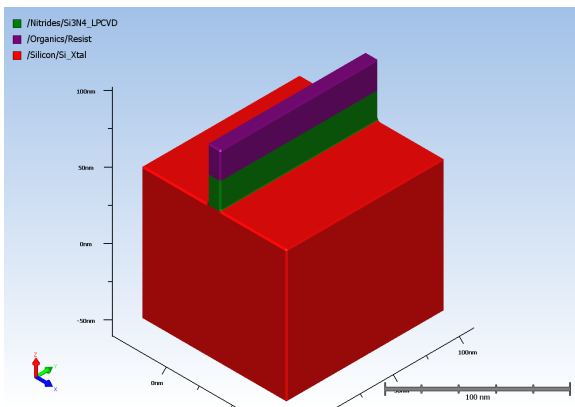


Figure 5: SiN hardmask Etch

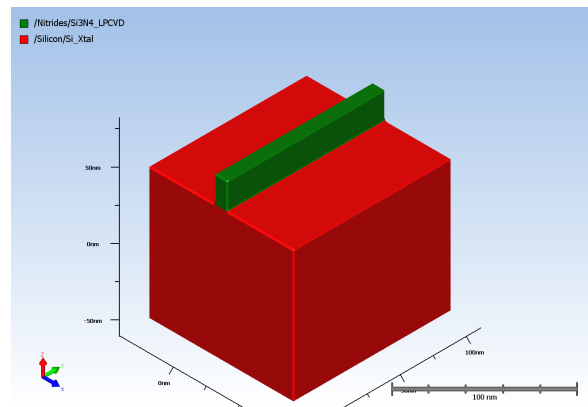


Figure 6: Resist removal

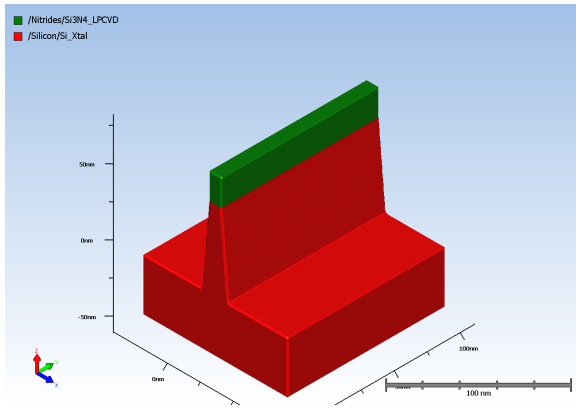


Figure 7: Fin Etch

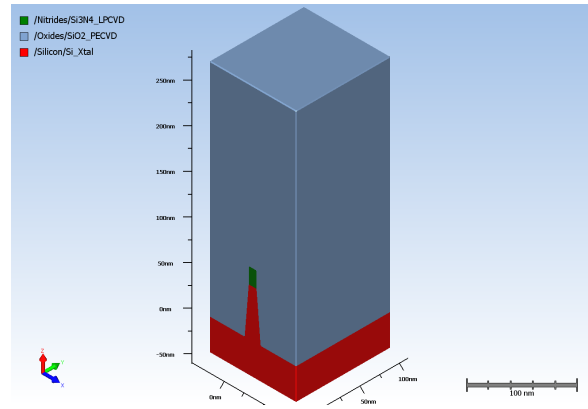


Figure 8: CVD deposit

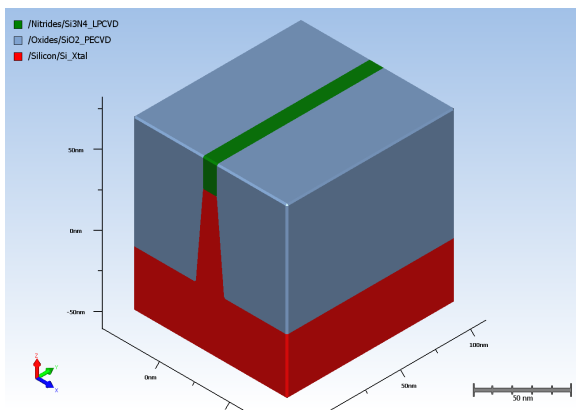


Figure 9: CMP

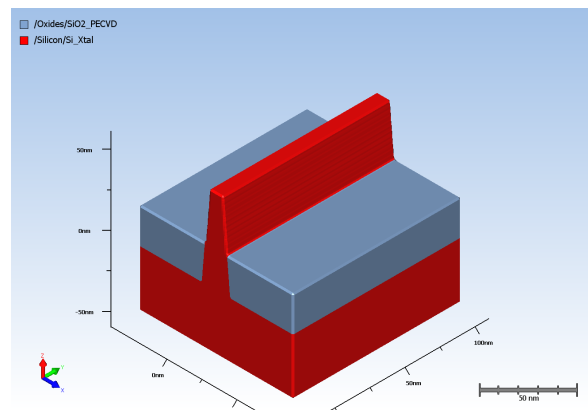


Figure 10: Nitride Oxide Etch

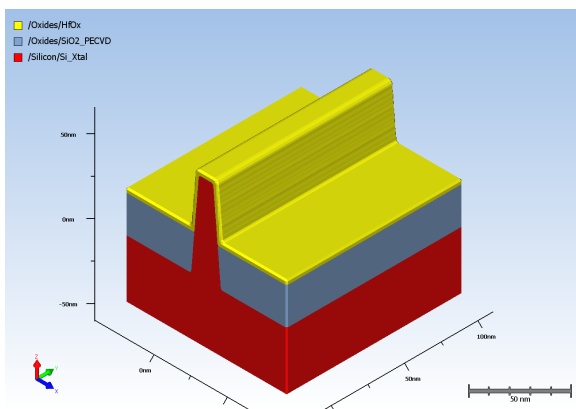


Figure 11: Gate High-K deposition

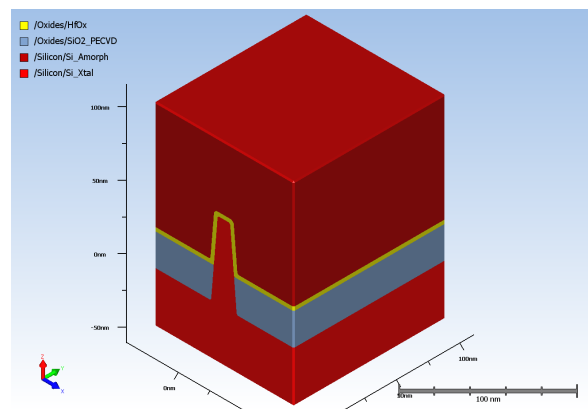


Figure 12: Gate poly Deposition

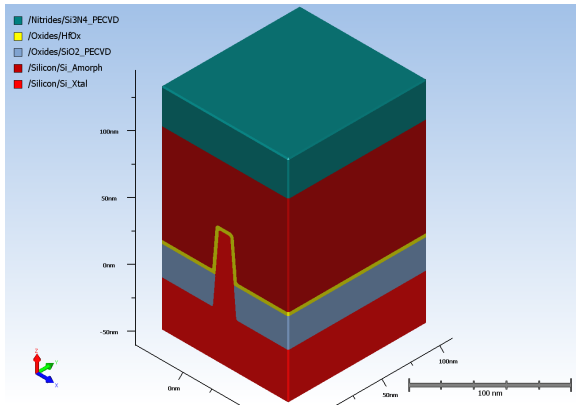


Figure 13: Gate Nitride deposition

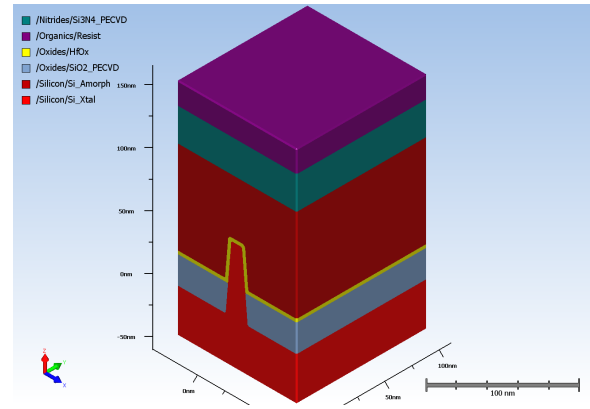


Figure 14: Gate poly Resist

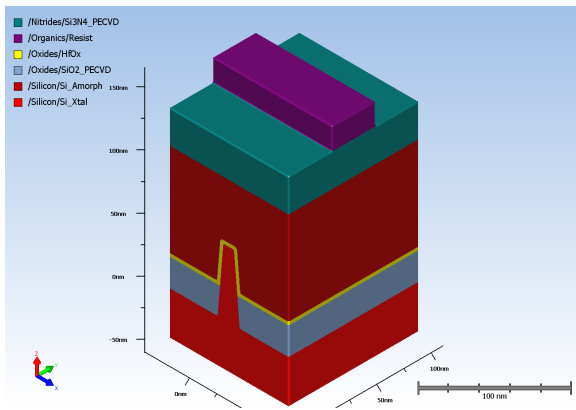


Figure 15: Expose material

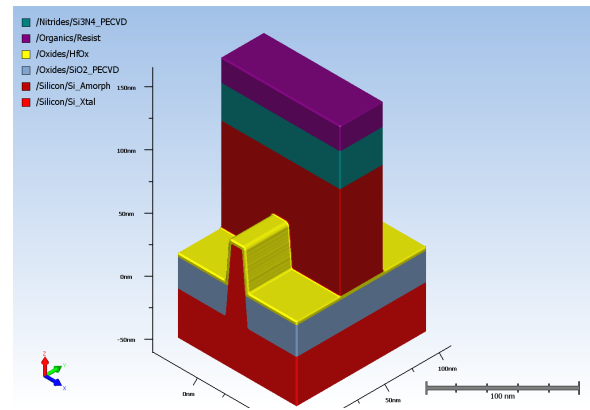


Figure 16: Nitride poly Etch

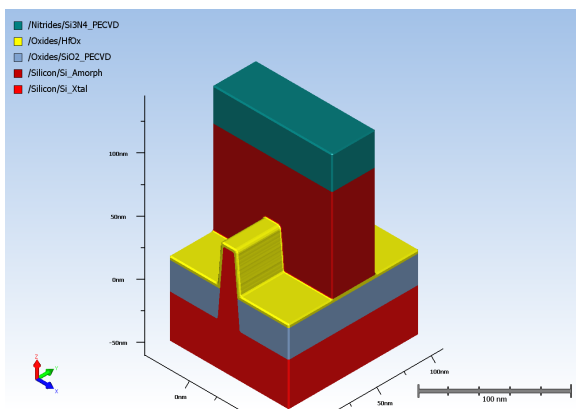


Figure 17: Resist Removal

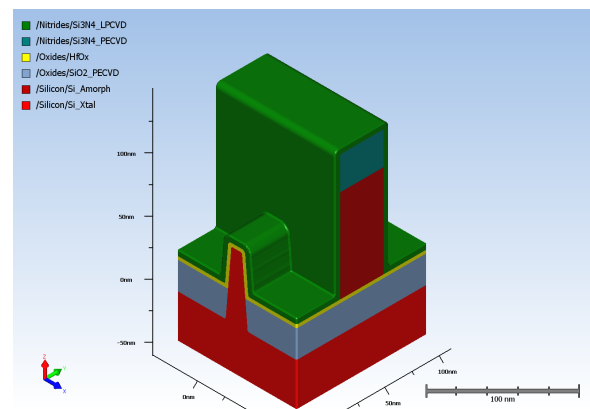


Figure 18: Spacer deposit

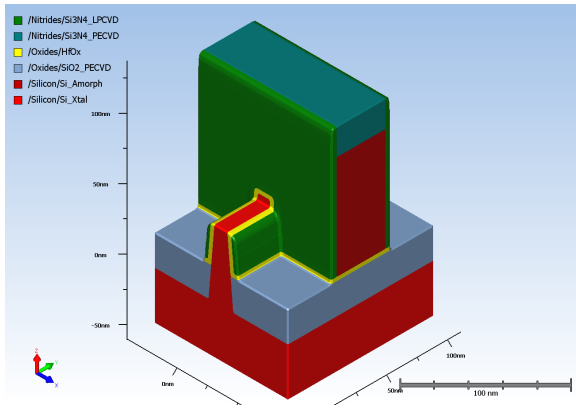


Figure 19: Spacer Etch 1

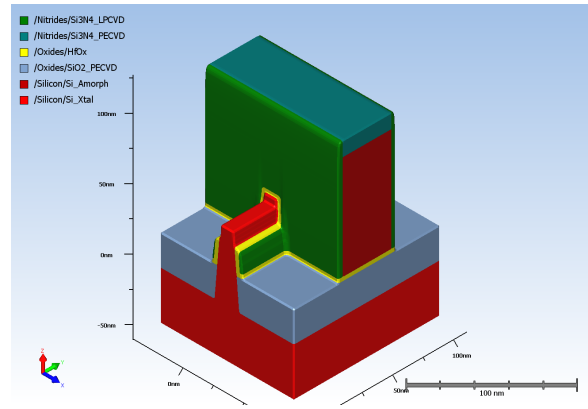


Figure 20: Spacer Etch 2

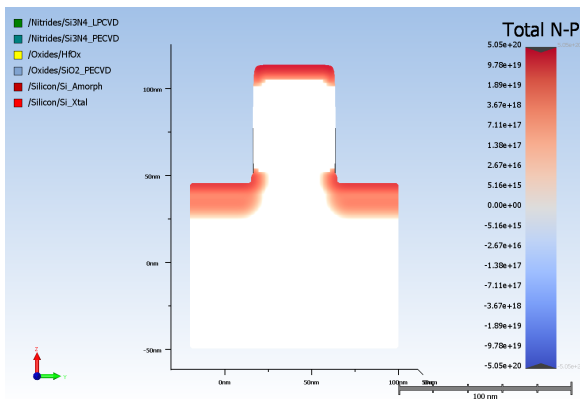


Figure 21: Implantation and Diffusion

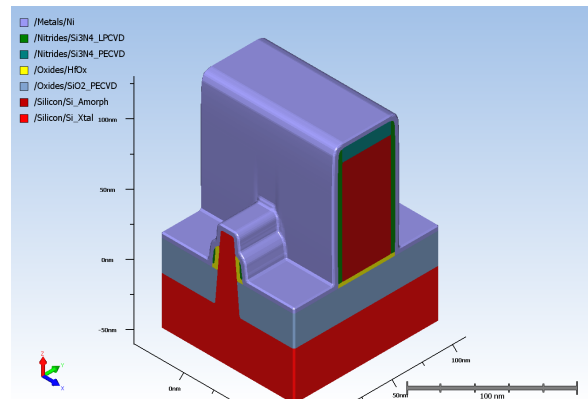


Figure 22: Ni Deposition

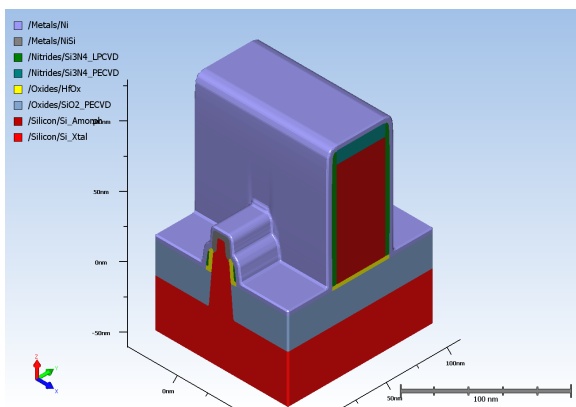


Figure 23: NiSi Transformation

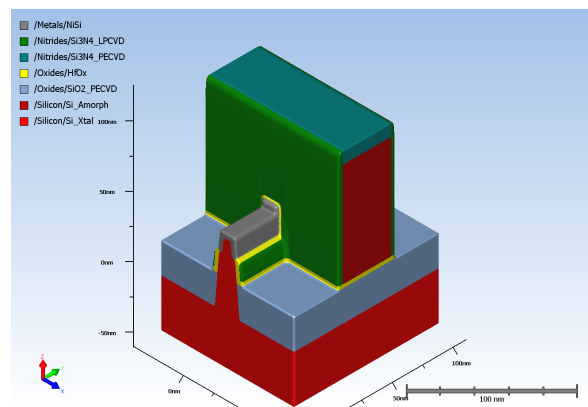


Figure 24: Removing Ni

Process Steps											Step Properties	
Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Lateral Ratio	Wafer	Name	Value
1		Wafer Setup	Wafer Setup	/Silicon/Si_Xtal	/PType/B	100				silicon wafer	Step Name	Implant
2		Sequence	Sequence								Action	Implant (Advanced)
2.1		Conformal Deposit	Si3N4 hardmask	/Nitrides/Si3N4...		20			1.0	silicon wafer	▼ Skp this ste Mask h required Skp this ste Wafer silicon wafer Wafer Side Top Dopant /NType/As Dose per T 14.5e13 Tilt Angle 8.5 ▼ Twist Angle + 0 X 0 Implant typ Analytical Implan /Silicon/Si_Xtal	
2.2		Planarizing Deposit	resist for Si3N4 hard mask	/Organics/Resist		20				silicon wafer		
2.3		Expose Material	Expose Resist	/Organics/Resist		0	Fin_Pattern	Light		silicon wafer		
2.4		Enhanced Basic Etch	Si3N4 hard mask Etch	/Nitrides/		20			0	silicon wafer		
2.5		Remove Materials	Resist Removal	/Organics/Resist						silicon wafer		
2.6		Taper Etch	Etch for Fin	/Silicon/Si_Xtal		60				silicon wafer	Comment - Implant	
2.7		Critical Dimension Metrology (A...	Metrology				Gate			silicon wafer		
3		Sequence	Sequence									
3.1		Planarizing Deposit	Planar Deposit	/Oxides/SiO2_P...		200				silicon wafer		
3.2		Polish	CMP							silicon wafer		
3.3		Enhanced Basic Etch	Nitride & Oxide Etch	/Nitrides/Si3N4...		55			0	silicon wafer	▼ Skp this ste Mask h required Skp this ste Wafer silicon wafer Wafer Side Top Dopant /NType/As Dose per T 14.5e13 Tilt Angle 8.5 ▼ Twist Angle + 0 X 0 Implant typ Analytical Implan /Silicon/Si_Xtal	
4		Sequence	Sequence									
4.1		Conformal Deposit	Gate High K Deposition	/Oxides/HfOx		3			1	silicon wafer		
4.2		Planarizing Deposit	Gate Poly Deposition	/Silicon/Si_Amo...		50				silicon wafer		
4.3		Planarizing Deposit	Gate Nitride Deposition	/Nitrides/Si3N4...		30				silicon wafer		
5		Sequence	Sequence								Comment - Implant	
5.1		Planarizing Deposit	Resist for Gate Poly	/Organics/Resist		20				silicon wafer		
5.2		Expose Material	Expose Material	/Organics/Resist		0	Gate	Light		silicon wafer		
5.3		Enhanced Basic Etch	Nitride & Poly Etch	/Nitrides/Si3N4...		115			0	silicon wafer		
5.4		Remove Materials	Resist Removal	/Organics/Resist						silicon wafer		
6		Sequence	Sequence								Comment - Implant	
6.1		Conformal Deposit	Spacer Deposit	/Nitrides/Si3N4...		5			1.0	silicon wafer		
6.2		Enhanced Basic Etch	Spacer Etch	/Nitrides/, /Oxi...		13			0.05	silicon wafer		
6.3		Enhanced Basic Etch	Spacer Etch	/Nitrides/, /Oxi...		12			0.05	silicon wafer		
7		Sequence	Sequence									
7.1		Implant (Advanced Modeling)	Implant		/NType/As					silicon wafer	Comment - Implant	
7.2		Diffusion (Advanced Modeling)	Diffusion	/Silicon/Si_Xtal	/NType/As					silicon wafer		
8		Sequence	Sequence									
8.1		Conformal Deposit	Ni Deposition	/Metals/Ni		4			1	silicon wafer		
8.2		Interface Growth	NiSi Transformation	/Metals/Ni, /Sili...						silicon wafer		
8.3		Replace Materials	Removing Ni	/Metals/Ni						silicon wafer		

Figure 25: Process Steps

Question 4

Modify the layout and follow similar process steps to create another fin of similar height and width separated by a distance of 50 nm.

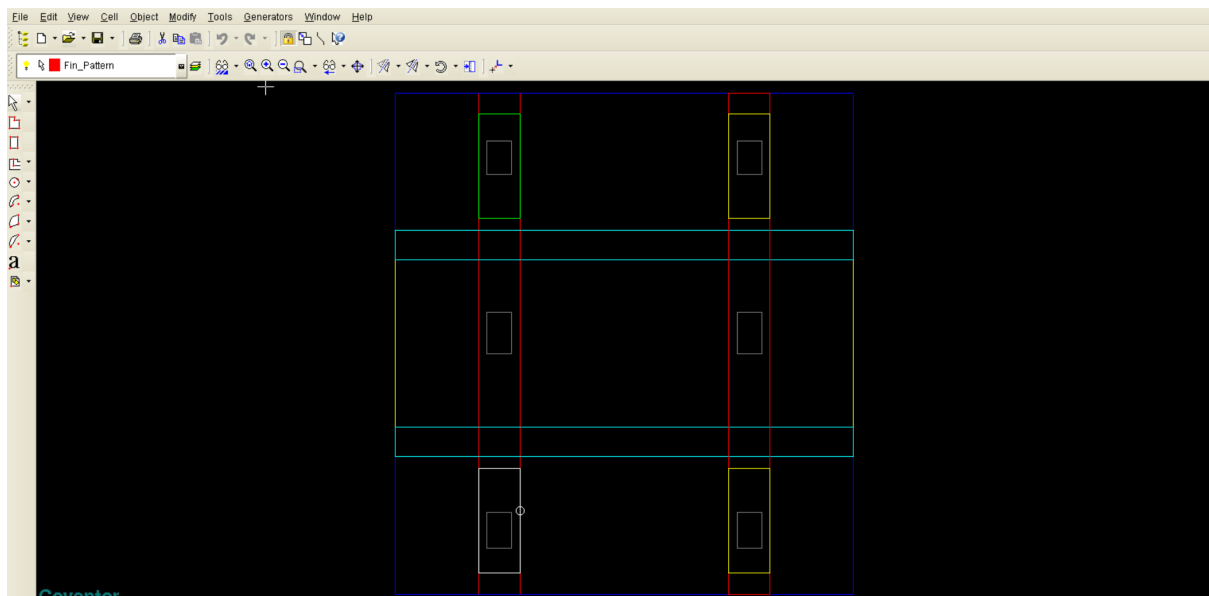


Figure 26: Finfet layout with two fins

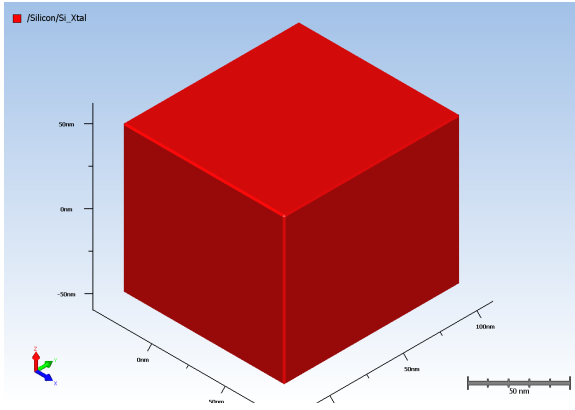


Figure 27: Wafer Setup

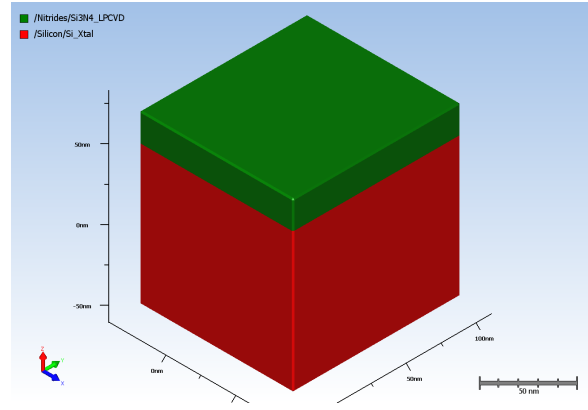


Figure 28: SiN Hardmask

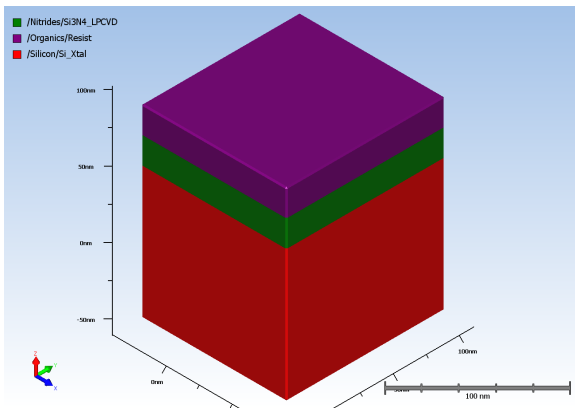


Figure 29: Resist for SiN

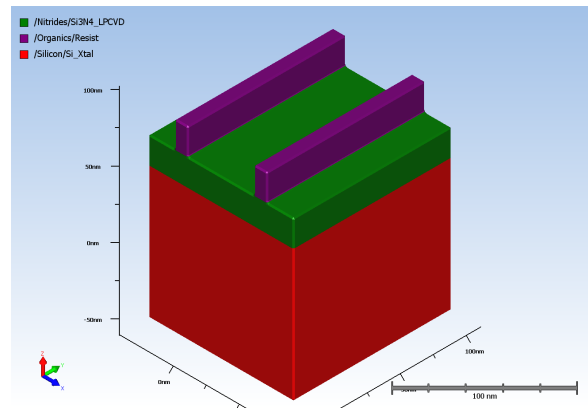


Figure 30: Expose resist

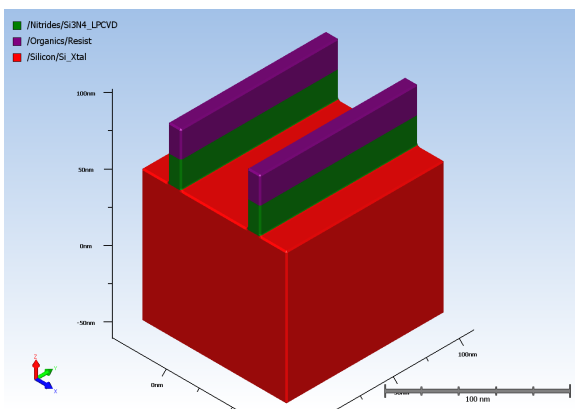


Figure 31: SiN hardmask Etch

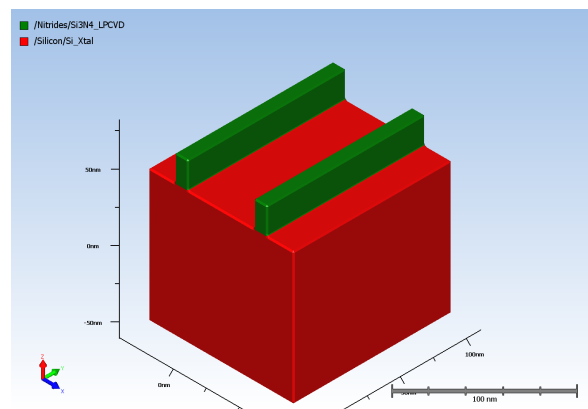


Figure 32: Resist removal

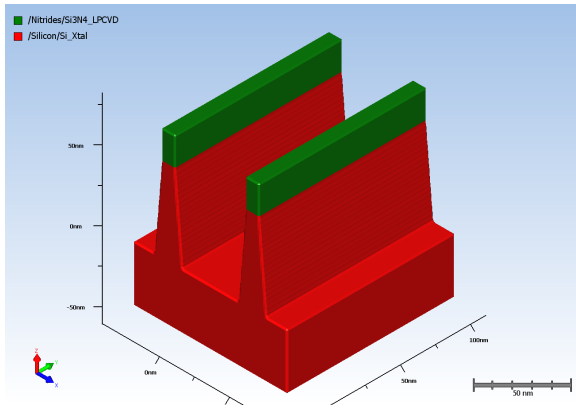


Figure 33: Fin Etch

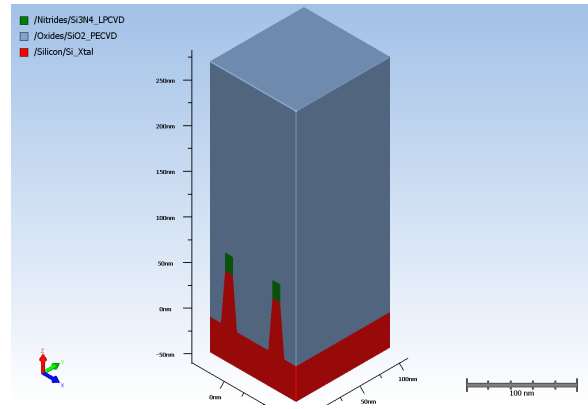


Figure 34: CVD deposit

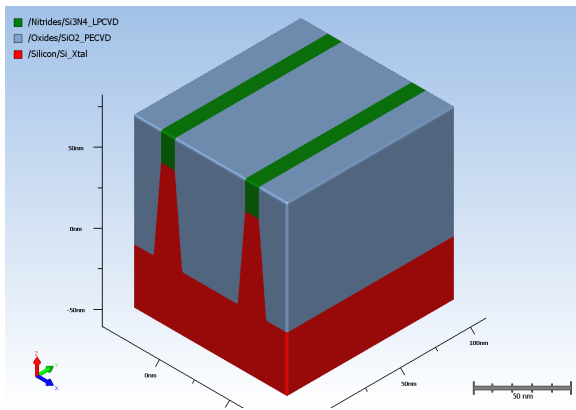


Figure 35: CMP

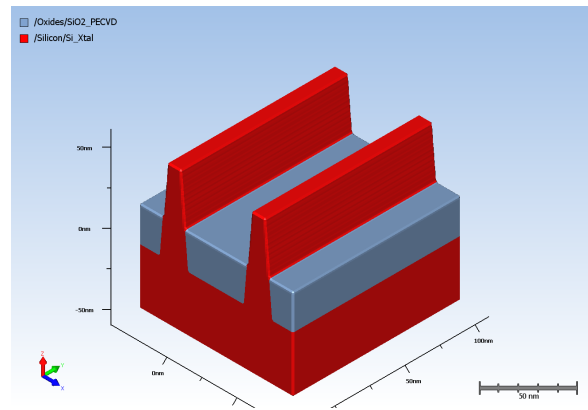


Figure 36: Nitride Oxide Etch

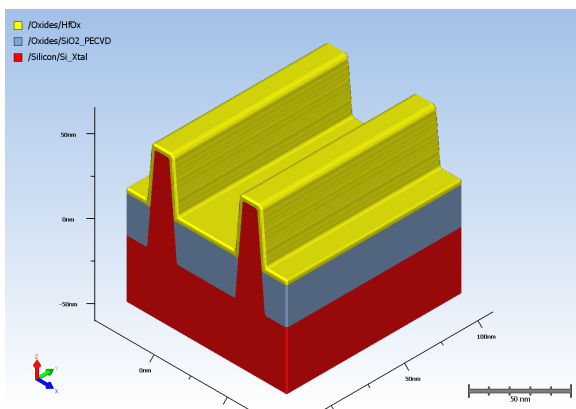


Figure 37: Gate High-K deposition

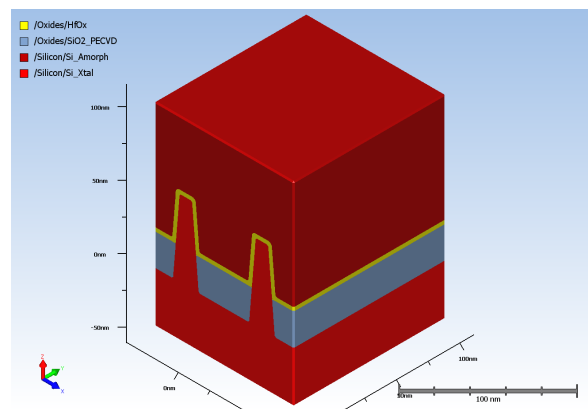


Figure 38: Gate poly Deposition

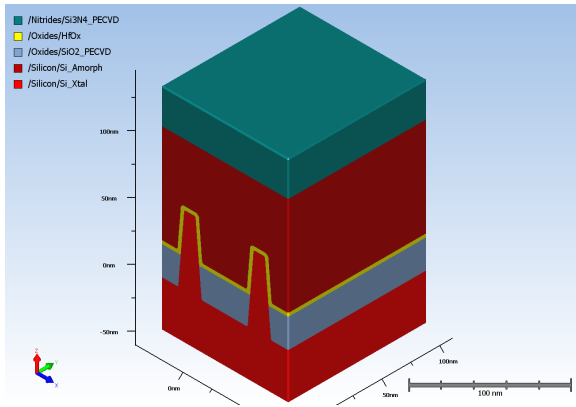


Figure 39: Gate Nitride deposition

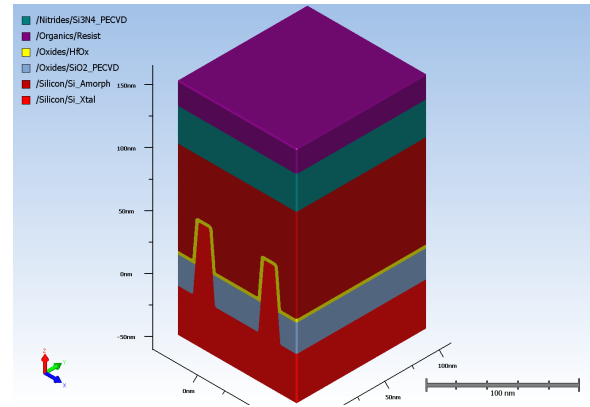


Figure 40: Gate poly Resist

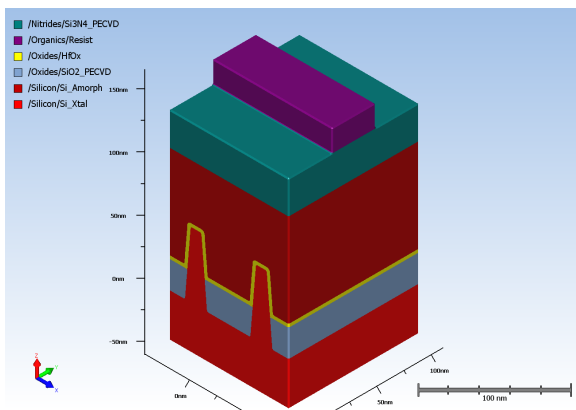


Figure 41: Expose material

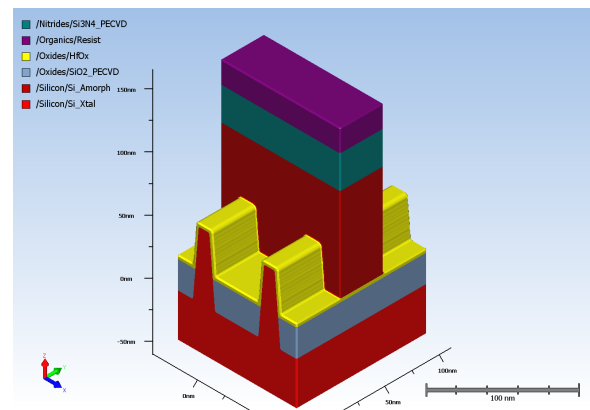


Figure 42: Nitride poly Etch

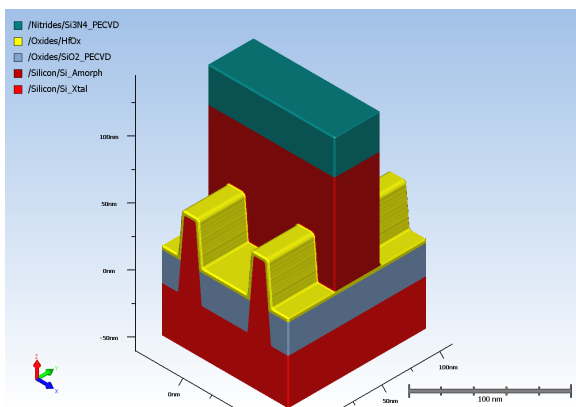


Figure 43: Resist Removal

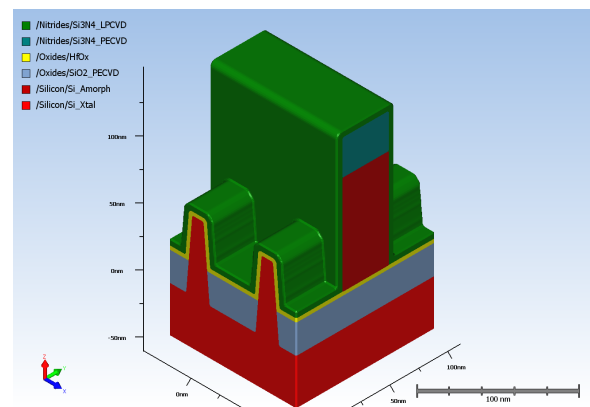


Figure 44: Spacer deposit

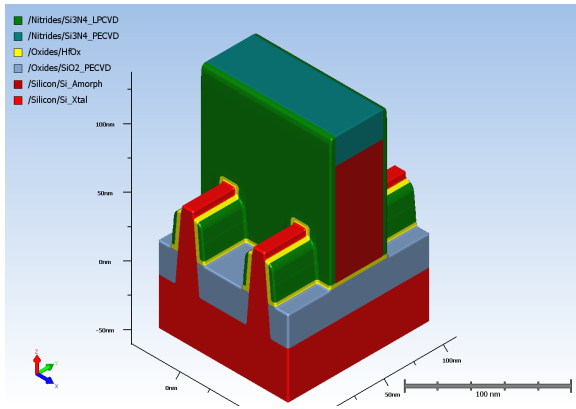


Figure 45: Spacer Etch 1

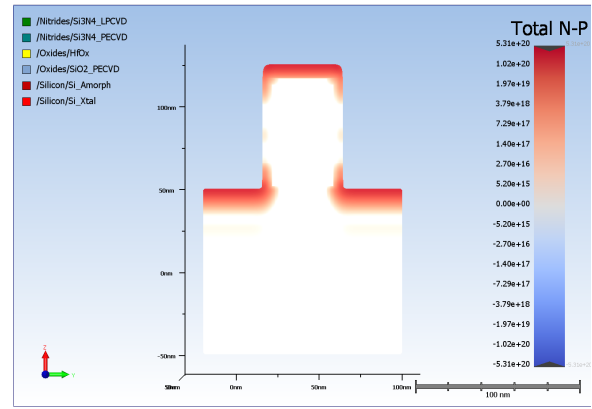


Figure 46: Implantation and Diffusion

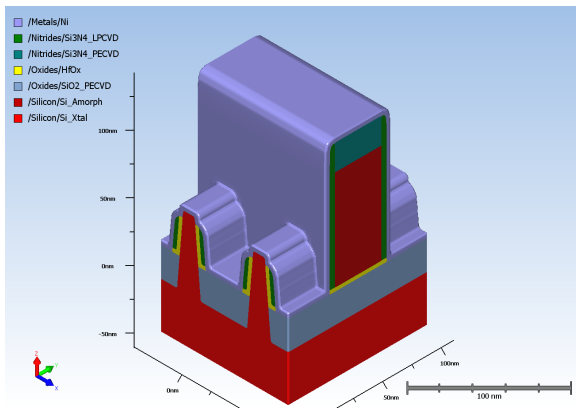


Figure 47: Ni Deposition

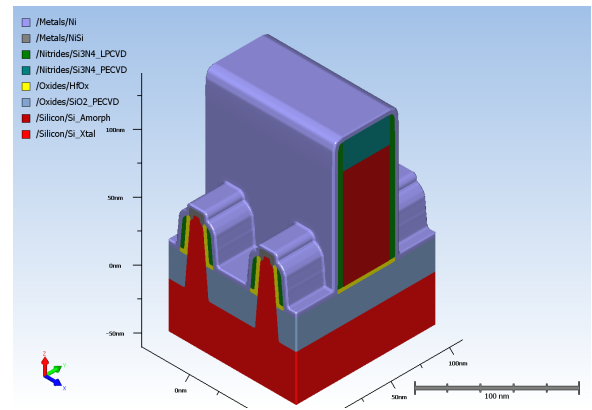


Figure 48: NiSi Transformation

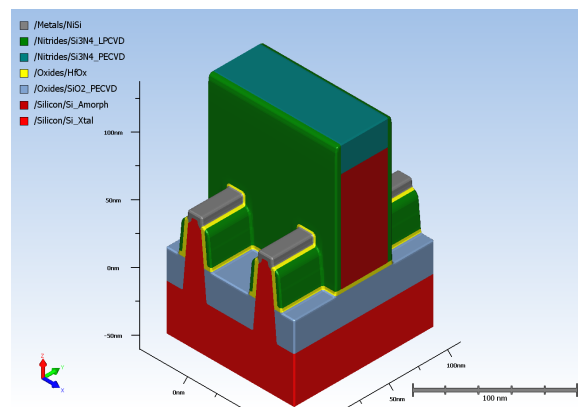


Figure 49: Removing Ni

Question 5

What is the purpose of salicidation in the described process flow? What benefits does salicidation provide compared to a direct metal-silicon contact?

Salicidation (self-aligned silicide formation) is used to form a low-resistance contact on the exposed silicon regions such as the source, drain, and gate after implantation and activation. The primary purpose of salicidation is to reduce the parasitic resistance at these regions and to create a stable, uniform metal–silicon interface.

Purpose of Salicidation:

- To reduce contact resistance at the source/drain and gate.
- To improve the drive current by lowering parasitic resistance.
- To provide a stable and uniform metal–silicon interface.
- To prevent uncontrolled metal diffusion into silicon by forming a controlled silicide layer.

Benefits of Salicidation Compared to Direct Metal–Silicon Contact:

- Lower contact resistance due to silicide formation, unlike the high Schottky barrier in direct metal–silicon contact.
- Improved thermal stability, as silicides form a stable interface that prevents excessive metal diffusion.
- Better uniformity and self-aligned behavior with gate spacers and fin structures.
- Controlled silicon consumption during silicide formation.
- Lower leakage currents and improved device reliability.